Project report on

Brain Tumor Detection

A Dissertation submitted in partial fulfilment of the Academic requirements for the award of the degree.

Bachelor of Technology

In

Computer Science and Engineering

Submitted by

Gadipe Shivani
18H51A0510

Under the esteemed guidance of
Mr. M. Shiva Kumar
Assistant Professor Dept. of CSE

Department of Computer Science and Engineering

CMR COLLEGE OF ENGINEERING & TECHNOLOGY

(An Autonomous Institution under UGC & JNTUH , Approved by AICTE, Permanently Affiliated to JNTUH, Accredited by NBA.)

KANDLAKOYA, MEDCHAL ROAD, HYDERABAD – 501401

2018- 2022

Project report on

Brain Tumor Detection

A Dissertation submitted to JNTU Hyderabad in partial fulfillment of the academic requirements for the award of the degree.

Bachelor of Technology

In

Computer Science and Engineering

Submitted by

Gadipe Shivani
18H51A0510

Under the esteemed guidance of
Mr. M. Shiva Kumar
Assistant Professor Dept. of CSE

EXPLORE TO INVENT

Department of Computer Science and Engineering

CMR COLLEGE OF ENGINEERING & TECHNOLOGY

(An Autonomous Institution under UGC & JNTUH , Approved by AICTE, Permanently Affiliated to JNTUH, Accredited by NBA.)

KANDLAKOYA, MEDCHAL ROAD, HYDERABAD – 501401

2018- 2022

CMR COLLEGE OF ENGINEERING & TECHNOLOGY

KANDLAKOYA, MEDCHAL ROAD, HYDERABAD – 501401

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Major Project report entitled "**Brain Tumor Detection**" being submitted by **Gadipe Shivani** ***(18H51A0510)*** in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering is a record of bonafide work carried out her under my guidance and supervision.

The results embodies in this project report have not been submitted to any other University or Institute for the award of any Degree.

Mr. M. Shiva Kumar

Assistant Professor
Dept. of CSE

Dr. K Vijaya kumar

Professor and HOD
Dept. of CSE

ACKNOWLEDGEMENT

With great pleasure I want to take this opportunity to express my heartfelt gratitude to all the people who helped in making this project work a grand success.

I am grateful to **Mr. M. Shiva Kumar**, Assistant Professor , Dept of Computer Science and Engineering for his valuable suggestions and guidance during the execution of this project work.

I would like to thank **Dr. Vijaya Kumar Koppula**, Head of the Department of Computer Science and Engineering, for his moral support throughout the period of my study in CMRCET.

I am highly indebted to **Dr. V A Narayana**, Principal CMRCET for giving permission to carry out this project in a successful and fruitful way.

I would like to thank the Teaching & Non- teaching staff of Department of Computer Science and Engineering for their co-operation

Finally I express my sincere thanks to **Mr. Ch. Gopal Reddy**, Secretary, CMR Group of Institutions, for his continuous care. I sincerely acknowledge and thank all those who gave support directly and indirectly in completion of this project work.

Gadipe Shivani

(18H51A0510)

TABLE OF CONTENTS

CHAPTER NO.		TITLE	PAGE No.
		LIST OF FIGURES	I
		ABSTRACT	II
1		INTRODUCTION	
	1.1	Introduction to Brain Tumor	2-3
	1.2	Types of Brain Tumors	3-6
	1.3	Motivation	6-8
	1.4	Problem Definition	8
	1.5	Objectives	8
2		BACKGROUND WORK	
	2.1	Domain Introduction	10-11
	2.2	Model Description	11-12
	2.3	Existing Systems	13-20
		2.3.1 Brain Tumor Classification using Support Vector Machine	13-15
		2.3.2 Brain Tumor Detection Using Image Processing	15-18
		2.3.3 Tumor Region Extraction Using Edge Detection Method In Brain MRI Image	18-20
	2.4	Comparison of Existing Models	21-22
3		BRAIN TUMOR DETECTION USING UNET ARCHITECTURE	

	3.1	Introduction	24
	3.2	Dataset Description	24-25
	3.3	UNet Architecture	25-26
	3.4	Proposed System Implementation	26-30
	3.5	Architecture of Proposed System	31
	3.6	Hardware and Software Requirements	31
4		DESIGNING	
	4.1	Introduction to UML Diagrams	33-34
	4.2	Use case Diagram	35
	4.3	Sequence Diagram	36
	4.4	Activity Diagram	37
	4.5	Class Diagram	38
5		RESULT AND DISCUSSION, PERFORMANCE ANALYSIS	
	5.1	Performance Analysis	40-41
	5.2	Output	42-44
6		SUMMARY AND CONCLUSION	
	6.1	Summary	46
	6.2	Conclusion	47
7		REFERENCE	49

List of Figures

FIGURE NO.	TITLE	PAGE No.
1.1	Sample Image of Brain Tumor	2
1.2	Image of Glioma	4
1.3	Image of Glioblastoma	4
1.4	Image of Ependymoma	5
1.5	Image of Astrocytoma	6
1.6	Image of Brain Metastases	6
1.7	Tumor location analysis	7
1.8	CNN Architecture	10
1.9	Flow diagram of existing system- 1	14
2.0	Resultant image of existing system-1	15
2.1	Flow diagram of existing system- 2	16
2.2	Resultant image of existing system-2	17
2.3	Flow diagram of existing system-3	19
2.4	Resultant image of existing system-3	20
2.5	Architecture of UNet	26
2.6	Flow of the project	30
2.7	Architecture of Project	31
2.8	Types of UML Diagram	34
2.9	Usecase Diagram	35
3.0	Sequence Diagram	36
3.1	Activity Diagram	37
3.2	Class Diagram	38
3.3	Training the MRI images	40

3.4	**Masking images**	**40**
3.5	**Comparing Accuracy and Loss of Model**	**41**
3.6	**Image of the Web Page**	**42**
3.7	**Image showing how to choose MRI Image**	**43**
3.8	**Image showing the file name**	**43**
3.9	**Result of Predicted mask image**	**44**

ABSTRACT

Detecting a brain tumor at an early stage is a challenging undertaking for doctors. Images from an MRI or CT scan can be used to detect a tumor. These MRI pictures are more susceptible to image noise and other environmental influence than regular MRI images. The image quality is affected by noise in this case. The picture quality degrades, making it impossible to extract the characteristics and causes of the tumor from the image. So, automatic and reliable segmentation methods are required; however, the large spatial and structural variability among brain tumors make automatic segmentation a challenging problem. In this paper, we propose an automatic segmentation method based on Convolutional Neural Networks (CNN), exploring small 3*3 kernels. The use of small kernels allows designing a deeper architecture, besides having a positive effect against over fitting, given the fewer number of weights in the network. We also investigated the use of intensity normalization as a pre-processing step, which though not common in CNN-based segmentation methods, proved together with data augmentation to be very effective for brain tumor segmentation in MRI images.

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION TO BRAIN TUMOR:

Tumors are a large family of diseases that involve abnormal cell growth with the potential to invade or spread to other parts of the body. They form a subset of neoplasms. A neoplasm or tumor is a group of cells that have undergone unregulated growth and will often form a mass or lump, but may be distributed diffusely. All tumor cells show the six hallmarks of cancer. These characteristics are required to produce a malignant tumor. They include:

- Cell growth and division absent the proper signals.
- Continuous growth and division even given contrary signals.
- Avoidance of programmed cell death.
- Limitless number of cell divisions.
- Promoting blood vessel construction.

Fig: 1.1- Sample Image of Brain Tumor

The progression from normal cells to cells that can form a detectable mass to outright tumor involves multiple steps known as malignant progression. Brain Tumor is the term used to describe the growth of abnormal cells lining over the brain. These cells divide and grow more rapidly than normal cells, which would combine to form a cluster or tumor. These days brain tumor has become one of the major causes for the death. Brain tumor has the highest

mortality rate of all the cancers in this world. Survival from brain tumor depends on the tumor size at the time of detection. Brian Tumor can be divided into two main groups, non-small Brain tumor and small cell brain tumor. These assigned types of brain tumors depend on their characteristics. In general, there are four types of brain tumors i.e., I to IV. Staging can be done based on the size of the tumor.

According to the statistics conducted by the World Health Organization (WHO), cancer is the leading cause of death, accounting for an estimated 9.6 million deaths in 2018. Tumor tops the list with an estimate of 2.09 million cases in which 1.76 million people have died. The death rate due to brain tumor forms 21% of all cancer deaths. In the year 2019, there are 228,150 new cases estimated out of which 142,670 are estimated to die which forms 62% of the death rate. Moreover, the death rates of brain tumors are expected to increase and which will wind up to 21 million at the end of 2032.

1.2 TYPES OF BRAIN TUMORS:

a) Glioma:

A glioma is a type of tumor that starts in the glial cells. The glial cells surround and support the neurons in the brain and other parts of the nervous system. Gliomas are one of the most common types of primary brain tumors. They can be low grade (slow growing) or high grade (fast growing). Sometimes low-grade gliomas change into high-grade gliomas over time.

The symptoms of a glioma can vary depending on the type of tumor you have and where it is located. These can include headache, seizures, nausea and vomiting, problems speaking or performing everyday tasks, and personality and mood changes.

Fig: 1.2- Image of Glioma

b) Glioblastoma:

Glioblastoma is the most aggressive type of brain cancer in adults. The tumors start in astrocyte cells, which support your brain's nerve cells in various ways. Glioblastomas can metastasize (spread) through brain tissue and can be difficult to remove. In recent years, doctors at MSK and elsewhere have begun to recognize distinct molecular subtypes of glioblastomas, which may in turn enable us to find new and targeted treatment approaches.

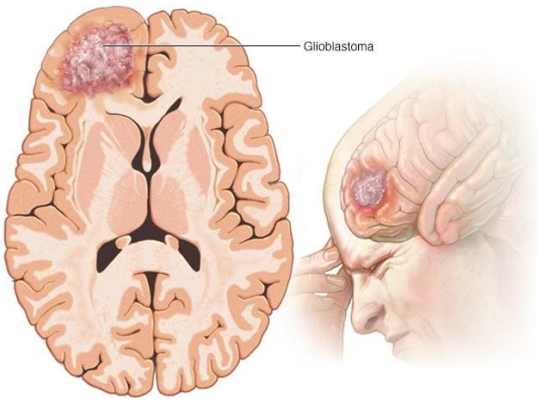


Fig: 1.3- Image of Glioblastoma

c) Ependymoma:

An ependymoma is a primary central nervous system (CNS) tumor. This means it begins in the brain or spinal cord. To get an accurate diagnosis, a piece of tumor tissue will be removed during surgery, if possible. A neuropathologist should then review the tumor tissue. Symptoms related to an ependymoma depend on the tumor's location. Here are some possible symptoms that can occur. People with an ependymoma in the brain may have: Headaches, Nausea, Vomiting, Dizziness.

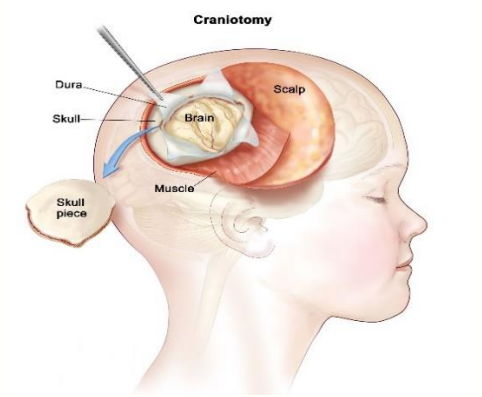


Fig: 1.4- Image of Ependymoma

d) Astrocytoma:

Astrocytoma is a type of cancer that can form in the brain or spinal cord. Astrocytoma begins in cells called astrocytes that support nerve cells. Astrocytoma signs and symptoms depend on the location of your tumour. Astrocytoma as that occur in the brain can cause seizures, headaches and nausea. Astrocytoma as that occur in the spinal cord can cause weakness and disability in the area affected by the growing tumour. Astrocytoma can be a slow-growing tumour, or it can be an aggressive cancer that grows quickly. The aggressiveness (grade) of your astrocytoma determines your prognosis and treatment options.

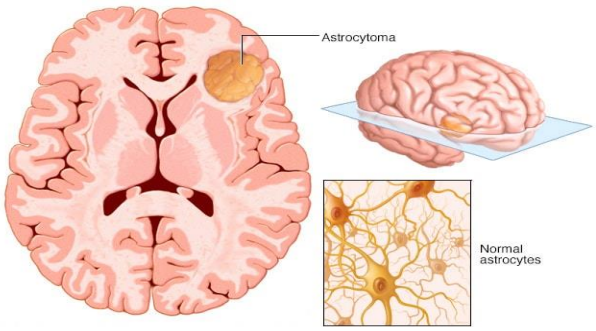


Fig: 1.5- Image of Astrocytoma

e) Brain Metastases:

Brain metastases occur when cancer cells spread from their original site to the brain. Any cancer can spread to the brain, but the types most likely to cause brain metastases are lung, breast, colon, kidney and melanoma. Brain metastases may form one tumor or many tumors in the brain. As the metastatic brain tumors grow, they create pressure on and change the function of surrounding brain tissue. This causes signs and symptoms, such as headache, personality changes, memory loss and seizures.

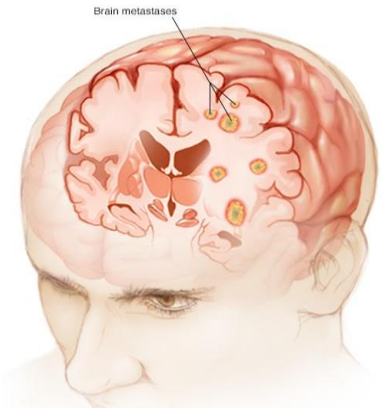


Fig: 1.6- Image of Brain Metastases

1.3 MOTIVATION:

Brain tumor is treated as a worldwide predicament and is a major reason behind the cancer related deaths in humans around the globe. It is one of the deadliest cancers with the least survival rate after diagnosis. The earlier the detection is, the higher the chances of successful treatment are. Brain tumor is caused by uncontrollable irregular growth of cells in brain tissue which proliferate in an uncontrolled way. It occurs after repeated inserts to the genetic material of the cell. Early detection of brain tissue abnormalities can help patients getting early treatment for their illness. Brain often has no symptoms until it has metastasized due to the limited number of pain receptors in the brain. The most frequently used imaging technique in the brain tumor diagnosis is Computed Tomography. Early detection of brain tumor, is an efficient way of improving the survival rate.

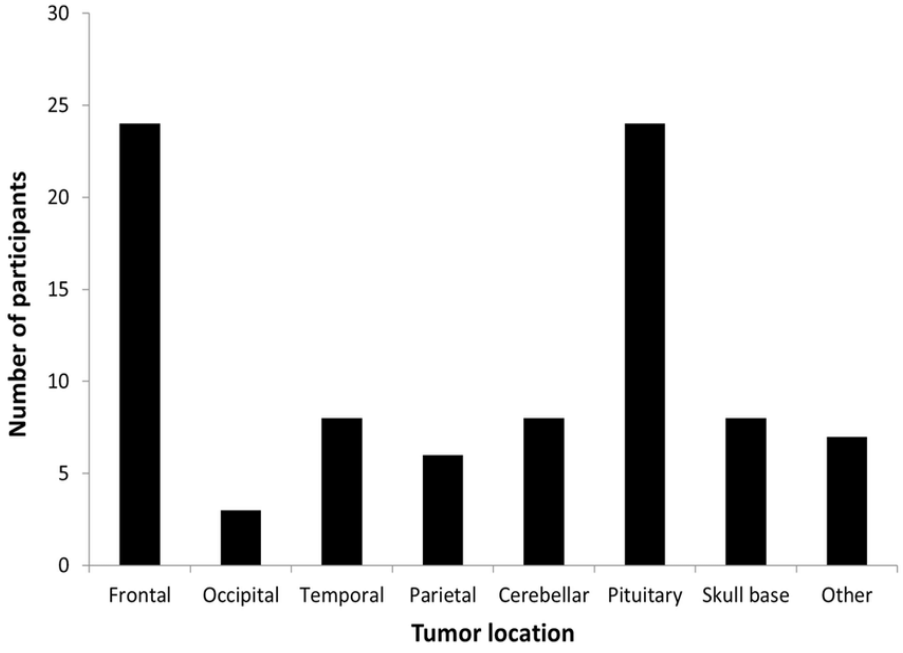


Fig: 1.7- Tumor location analysis

The motivation is to develop a software with better segmentation capability for use in medical imaging to detect diseases like brain tumor. Image segmentation has been identified as the key problem of medical image analysis and remains a popular and challenging area of research. Image segmentation is increasingly used in many clinical and research applications to analyze medical imaging datasets; which motivated us to present a snapshot of dynamically changing field of medical image segmentation.

CT (Computed Tomography), MRI (Magnetic Resonance Imaging etc. generates a large amount of image information. With the improved technology, not only does the size and resolution of the images grow but also the number of dimensions increases. In the future, we would like to have algorithms which can automatically detect diseases, lesions and tumors, and highlight their locations in the large pile of images. We have developed a system for detecting the tumor in the brain by using image processing techniques. This system can detect the tumor region based upon the area. The motivation of this work is to increase patient safety by providing better and more precise data for medical decision.

1.4 PROBLEM DEFINITION:

Brain tumor at early stage is very difficult task for doctors to identify. Tumor can be detected using MRI or CT scanned images. Considering the MRI images, due to the image noise and other environmental inferences the image quality gets damage. To overcome this types of issues we as a team decided to implement Brain Tumor Detection.

1.5 OBJECTIVES:

The main objective of our project is to detect the tumor in the brain whether the tumor is present or not, and specify the tumor location.

CHAPTER 2

BACKGROUND WORK

2.1 DOMAIN INTRODUCTION:

A Convolutional Neural Network, also known as CNN or ConvNet, is a class of neural networks that specializes in processing data that has a grid-like topology, such as an image. A digital image is a binary representation of visual data. It contains a series of pixels arranged in a grid-like fashion that contains pixel values to denote how bright and what color each pixel should be. The human brain processes a huge amount of information the second we see an image. Each neuron works in its own receptive field and is connected to other neurons in a way that they cover the entire visual field. Just as each neuron responds to stimuli only in the restricted region of the visual field called the receptive field in the biological vision system, each neuron in a CNN processes data only in its receptive field as well. The layers are arranged in such a way so that they detect simpler patterns first (lines, curves, etc.) and more complex patterns (faces, objects, etc.) further along. By using a CNN, one can enable sight to computers.

CNN Architecture:

A CNN typically has three layers: a convolutional layer, a pooling layer, and a fully connected layer.

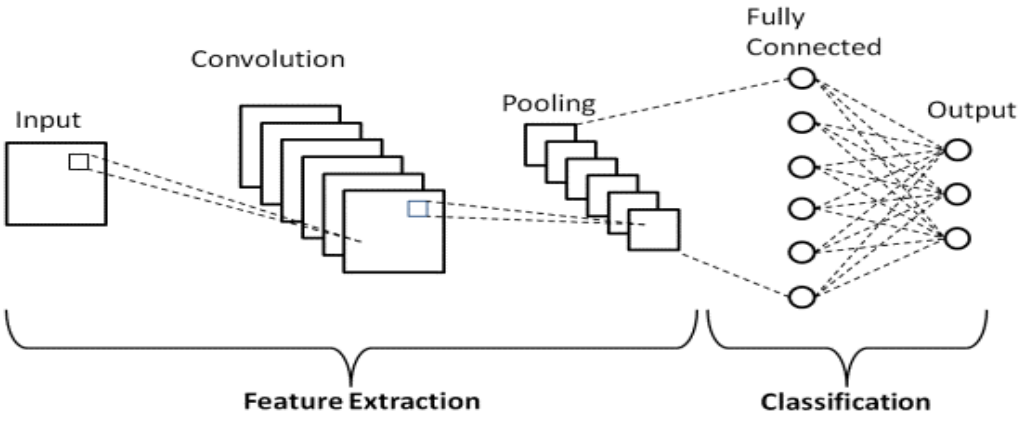


Fig: 1.8- CNN Architecture

A digital image is a binary representation of visual data. It contains a series of pixels arranged in a grid-like fashion that contains pixel values to denote how bright and what color each pixel should be.

2.2 MODEL DESCRIPTION:

Image Acquisition:

The first step of the MRI Image segmentation is image acquisition. High-quality MRI Image dataset used which are obtained from open Kaggle repository. The entire sample set is divided into two samples training and testing samples. Training sample consists of 80% data and the remaining images for testing.

Annotated Dataset Collection:

A Knowledge-based dataset is created by proper labeling of the collected images with unique classes.

Preprocessing:

The image is loaded as the .mat format and the image is in the form of MRI. MRI image is in either RGB or gray scale. The image is resized and converted to grey if the image was in color.

Feature extraction:

In this part, we extract features of images to use in classification part. For feature extraction and feature selection, we use convolution neural network (CNN). As previously described to classification the images in the first step, CNN extracts the features of the dataset and perform the classification based on these features. In KE-CNN method we use CNN for feature extraction. Our configuration for CNN is the same model used in the previous method for

convolutional neural networks. In this configuration, we have four convolution layers, four maxpooling layers with 1 fully connected layer. Fig.1 shows the block diagram KE-CNN algorithm.

Image segmentation:

It is a labelling process for each pixel in a medical image data set for indicating the type of tissue or the structure of anatomy. The labels that result from this approach have a vast area of applications in visualization and medical research. The method of dividing an image into sets of pixels which is also called super pixels is the image segmentation. The chief objective of segmentation is to identify the tumor's location. It is along chosen by the pixel powers themselves, yet in addition by the neighboring pixel powers and locations. Thought of these neighboring pixels extraordinarily controls the impact of noise.

Deployment:

The last step of machine learning life cycle is deployment, where we deploy the model in the real-world system. If the above-prepared model is producing an accurate result as per our requirement with acceptable speed, then we deploy the model in the real system. But before deploying the project, we will check whether it is improving its performance using available data or not. The deployment phase is similar to making the final report for a project.

UNet Architecture:

The UNet architecture was introduced for Biomedical Image segmentation. The introduced architecture had two main parts that were encoder and decoder. The encoder is all about the covenant layers followed by pooling operation. It is used to extract the factors in the image. The second part decoder uses transposed convolution to permit localization. It is again an F.C connected layers network.

2.3 EXISTING SYSTEMS:

2.3.1 Brain Tumor Classification using Support Vector Machine:

Author: N.Vani, A.Sowmya, N.Jayamma

Published Year: 2018

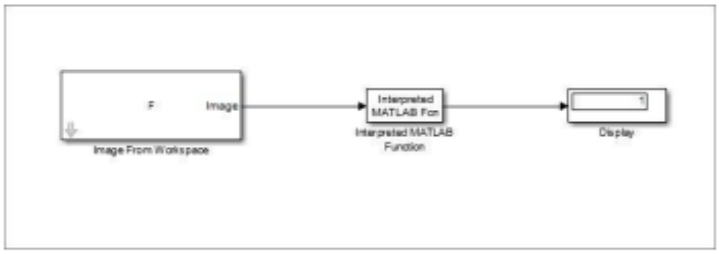
INTRODUCTION:

Object detection plays a major role in many areas like medical imaging, aerial surveillance, optimal manipulation and analysis, surgical microscopes, etc. The objective of this paper is to develop a model for brain tumors detection and classification i.e., to classify whether the tumor is cancerous or non-cancerous using SVM algorithm. Earlier many have detected using ANN which works on Empirical Risk Minimization. We are using Support Vector Machine algorithm that works on structural risk minimization to classify the images. The SVM algorithm is applied to medical images for the tumor extraction, and a Simulink model is developed for the tumor classification function. This paper presents a prototype for SVM-based object detection, which classifies the images and evaluates whether the classified image is cancerous or non-cancerous

IMPLEMENTATION:

The operation follows 5 steps namely: Image acquisition, pre-processing, K-means clustering, Morphological operation, tumor detection & area calculation.

SIMULINK MODEL FOR SVM CLASSIFICATION OF BRAIN TUMORS:



F is the matrix of the feature vectors i.e., 27X85 array that is given as input to the svm classification function. Moreover, the corresponding output of the svm classification. The results are shown in the command window of the MATLAB.

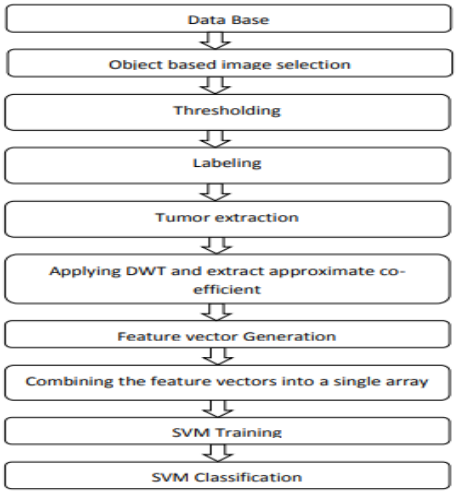


Fig: 1.9- Flow diagram of existing system- 1

CONCLUSION:

This paper presents an prototype for object detection with SVMs that can achieve real-time performance while maintaining high detection accuracies. 82% of accuracy is obtained and the positive predictive values (PPV) 81.48%, Negative predictive value (NPV) are calculated. The True positive cases are

22; True negative 5, False positive 5 and False negative are 22. Furthermore, the same prototype can be used for different application regardless of the window size, number of support vectors, and image size.

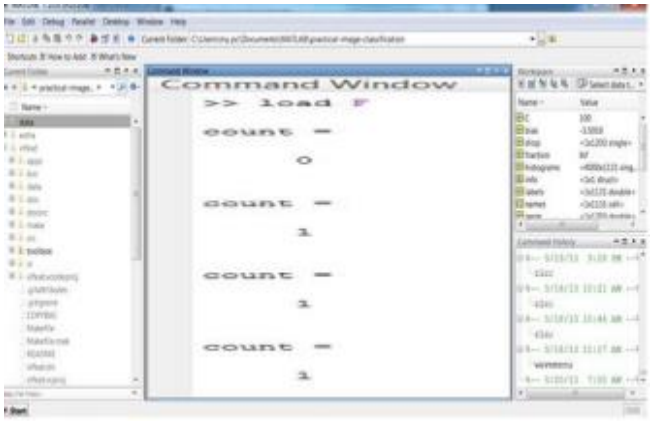


Fig: 2.0- Resultant image of existing syste-1

2.3.2 BRAIN TUMOR DETECTION USING IMAGE PROCESSING:

Author: Saurabh Kumar ,Iram Abid,Shubhi Garg

Published Year: 2019

INTRODUCTION:

This project proposes two different methodologies to segment a tumor from an MRI image and determine the type of tumor. For this one segmentation and one clustering techniques have been implemented. Each MRI image is passed through an imaging chain where the image is preprocessed to remove noise and is further enhanced to improve the contrast of the image. This paper proposes two different techniques which are then applied on the image to extract the tumor. These segmentation techniques include SOM Clustering and SVM Classification. Applying each of the segmentation techniques allows us to determine the most appropriate method to segment the tumor from each

of the images. The tumor region represents the pixel values for the foreground points extracted using the ginput() command from a texture image. The texture image is generated by applying the rangefilt() method. In order to enhance the texture characteristics of the image, smoothing filter is applied to the texture image. In this project, the major challenge faced was to locate and extract the proper tumor region from the image. Due to several lighting issues, unnecessary white portions were present in the image which could wrongly be segmented as a tumor. Also the unwanted noise and reduced contrast displays several regions from the image that are falsely claimed as a tumor. Another challenge faced was degraded quality of the MRI image due to several problems that would have occurred during the acquisition stage.

IMPLEMENTATION:

In this project, we have described our objective in two parts, the first half deals with detection of brain tumor that is the presence of the tumor in the provided MRI. The other part that is the second part contains the classification of the tumor. Here, we will analyze the MRI images which will conclude the stage of the tumor as benign or malignant. In general the diagram for our process. The input images will undergo various stages which can be summarized as follows that are shown in the figure.

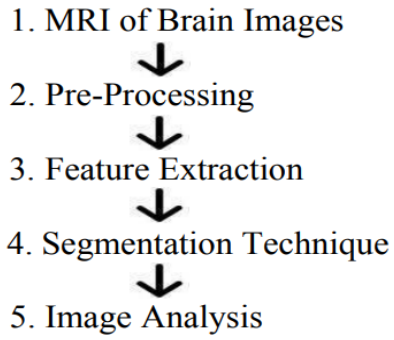


Fig: 2.1- Flow diagram of existing system- 2

CONCLUSION:

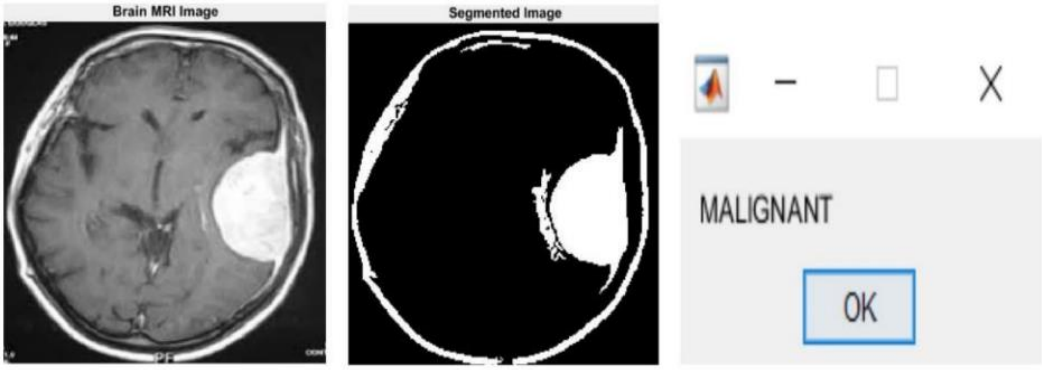


Fig: 2.2- Resultant image of existing syste-2

In this project they have automated the diagnosis procedure for the brain tumor detection by the use of image processing. Apart from several existing brain tumor segmentation and detection methodology are present for MRI of brain image our project has proved to provide an aver all accuracy by up to 97%. All the steps for detecting brain tumor that have been discussed starting from MRI image acquisition ,pre-processing steps to successfully classification of the tumor using the two segmentation techniques is been done.

Pre-processing involves operations like wavelet based methods has been discussed. Quality enhancement and filtering are important because edge sharpening, enhancement, noise removal and undesirable background removal are improved the image quality as well as the detection procedure. Among the different filtering technique, Gaussian filter suppressed the noise without blurring the edges and it is better outlier without reducing sharpness of the images. reduces the noise; enhance the image quality and computationally more efficient than other filtering methodology. After the image quality improvement and noise reduction discussed here, segmentation methodology for a brain tumor from MRI of brain image is been used. Classification based segmentation segment tumor accurately and

manufacture sensible results for big information set however undesirable behaviors can occur in case wherever a category is unrepresented in training data. Clustered based segmentation performs is straight forward, quick and manufacture sensible results for non-noise image except for noise pictures it leads to serious inaccuracy within the segmentation.

2.3.3 TUMOR REGION EXTRACTION USING EDGE DETECTION METHOD IN BRAIN MRI IMAGE:

Author: Gajendra Raut, Jyoti Bhagade

Published Year: 2020

INTRODUCTION:

Basically there are three types of brain tumor: Benign, Pre-malignant and Malignant. A benign tumor is one that does not grow abruptly. It never affects its neighboring tissues and not at all expand to other parts. eg. Moles. Pre-malignant tumor is supposed to be in precancerous stage and it can certainly lead to cancer if not properly treated at right time. Malignant tumor is one which worsens with the passing time and ultimately proves to be fatal. We can say malignancy is a tumor in descriptive or advance stage from where it is quite impossible to return back. Brain tumor is diagnosed by some tests like checking personal or family medical history, physical examination of body, brain angiogram, spinal biopsy etc. A patient can go under several imaging tests also eg. CT scan of brain where one can get detailed picture of brain.

IMPLEMENTATION:

The MRI scanned images are highly accurate to proceed with as here the computer is connected to a strong magnetic field which gives clear 2d picture

of the brain and located tumor in it. First of all we take an MRI image as input. If it is a color image, it is converted to gray scale image. The gray range is from 0-255, where 0 depicts white color and 255 shows black color. Gray conversion is essential as color image sometimes is not able to detect edges and some other features. Then we apply pre-processing step on it to remove any noise residing in our input image. Here median filter has been applied to remove the noise in input image. After that we will compute standard deviation of that filtered image. But it is not able to segment the tumor region in a proper way so we are compute the standard deviation. Then we calculate average intensity of the pixels that are above this standard deviation using the same deviated standard. And this computed average intensity is used as a threshold to proceed with the input MRI image. This will segment the tumor section. Finally, we will use sobel edge filter to get the tumor border region.

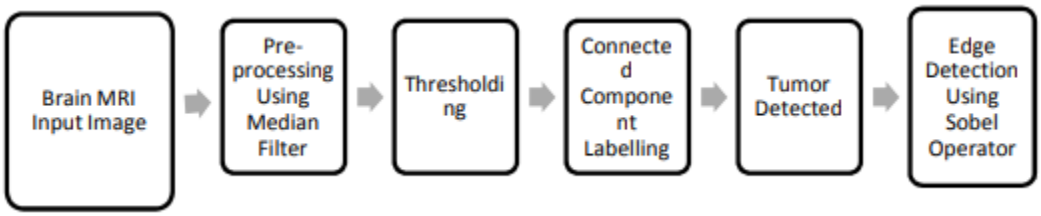


Fig: 2.3- Flow diagram of existing system-3

CONCLUSION:

We have proposed an automated method which can be an option for surgeons of brain tumor. We have used brain MRI images as our input image. It gives clear and robust 2d picture of the interior of skull which easily helps in diagnosing the tumor. We have used median filter and sobel edge detector for accurate and précised outcomes. Otherwise, the fault determination of the tumor will lead to mislead planning and treatment of the same. We must be cautious of reliability and accuracy as they are vital due to complexity residing in the tumor diagnosis.

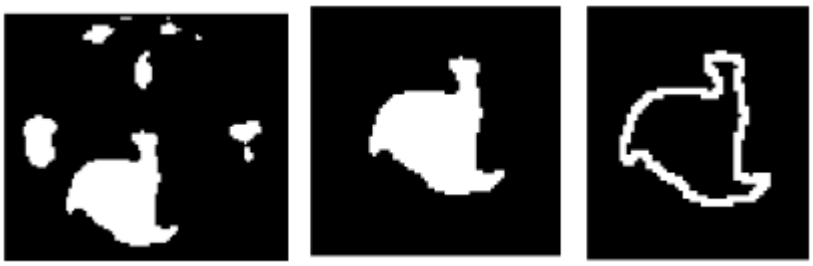


Fig: 2.4- Resultant image of existing syste-3

The automated method for brain tumor detection proposed above is able to meet the medical complexities regarding ongoing medical research in the field of tumor detection. The proposed method above finds the tumor region using Thresholding .We used here standard deviation for getting the threshold value but as it is dynamic we cannot get an appropriate value for the threshold. However we get the tumor region. But our output image contains so many unwanted objects along with the tumor.

For detecting tumor among these objects we find pixel area of all the objects in an image separately. Then we set a predefined area of 5oo pixels for the tumor extraction. If the area of any object is greater than 500 pixels, the resulting part is our tumor. We will discard any other object in an image which has a value smaller than this predefined area, saying it as unwanted object. Finally the tumor border will be identified by sobel gradient effectively in an efficient way. We have applied this proposed method on some 100 brain MRI images dataset where it gives same result each and every time. The exact detection of tumor in brain MRI opens a window of hope for millions of patients who are suffering from the same to get diagnosed well and sure. Above all, the proposed method has been implemented using MATLAB which is a powerful tool for image processing and mathematical functions.

2.4 COMPARISION OF EXISTING MODELS:

Support Vector Machine:

One of the best methods for classifying any image or pattern is SVM. SVM is used to split a set of images into two various classes. The classification is done by finding the hyper-plane that differentiate the two classes very well as given in Fig. One of the best methods for classifying any image or pattern is SVM. SVM is used to split a set of images into two various classes. The classification is done by finding the hyper-plane that differentiate the two classes very well as given in Fig. A support vector machine (SVM) is a supervised learning algorithm based on statistical learning theory. Given a labeled data set (training set), D= {|x,y||x -> data sample, y -> class label}, an SVM tries to compute a mapping function f such that f(x) = y for all samples in the data set. This mapping function describes the relationship between the data samples and their respective class labels; and is used to classify new unknown data. Classification in the context of SVMs is done using the following classification decision function (a process called the feed-forward phase).

Merits:

The method is more suitable for detecting brain tumours in high glioma images

Demerits:

Support Vector Machine failed to identify tumour images correctly for low glioma images

Image Processing:

Image processing is the process of transforming an image into a digital form and performing certain operations to get some useful information from it.

The image processing system usually treats all images as 2D signals when applying certain predetermined signal processing methods.

Merits:

Automatic detection of brain tumor through MRI can provide the valuable outlook and accuracy of earlier brain tumor detection.

Demerits:

This systems is confined only to segmentation

Edge Detection:

Edge detection is an image processing technique for finding the boundaries of objects within images. It works by detecting discontinuities in brightness. Edge detection is used for image segmentation and data extraction in areas such as image processing, computer vision, and machine vision.

Merits:

Tumor border is identified by sobel gradient effectively in an efficient way.

Demerits:

High computational complexity.

CHAPTER-3

BRAIN TUMOR DETECTION USING UNET ARCHITECTURE

3.1 INTRODUCTION TO PROPOSED SYSTEM:

Convolution Neural Network contains neurons with some weights and biases. These neurons receive some inputs from anterior layer. It calculates a dot product between the input and weights, and optionally follows it with a non-linearity. The CNN architectures makes postulation that the inputs are images which sanctions us to encode certain properties into the architecture. Convolution layer contain more than one convolution layers with activation function ReLU and pooling step. At the end final layer is plenary connected layer is a standard neural network architecture.

The proposed implementation includes dataset collection and description of the dataset, preprocessing the images, classification of brain tumors using CNN and performance measures for the system.

3.2 DATASET DESCRIPTION:

The brain MRI images are obtained from Kaggle Website. It has 766 images with the segmentation mask. The database is divided into training set and testing set. The training set contains 612 images out of which 80 % of data is used for training the network. The test dataset consists of 154 images used to test the developed model. The test dataset contains new images unseen in the training dataset to ensure robustness of the network. The carotid ultrasound images are obtained from radiology department of Apollo Hospitals, Chennai using Philips iU22 ultrasound machine. Ethical clearance was obtained for images acquired. A total of 644 images are obtained, out of which 44 images are used for testing the network and rest 600 images are used for training and validating the network in the ratio of 80:20. The chest x-rays are obtained from kaggle database. The database has 278 x-ray images out of which 225 images are used for training and validating the network. The rest 53 images are used for testing the network. The brain MRI images and

chest x-ray images obtained from repository are pre-processed images. The carotid ultrasound images obtained from Apollo Hospital are raw images and need to be pre-processed. The image contains details of vendor of ultrasound machine and machine model details termed as back ground of image. Graph cut algorithm is implied to extract useful carotid image from the background information details. The images obtained have intensity variation among them. Intensity matching is done to match the intensities. These preprocessed images are used by neural network for segmentation

3.3 UNET ARCHITECTURE:

The UNet architecture was introduced for Biomedical Image segmentation. The introduced architecture had two main parts that were encoder and decoder. The encoder is all about the covenant layers followed by pooling operation. It is used to extract the factors in the image. The second part decoder uses transposed convolution to permit localization. It is again an F.C connected layers network.

U-Net is an architecture for semantic segmentation. It consists of a contracting path and an expansive path. The contracting path follows the typical architecture of a convolutional network. It consists of the repeated application of two 3x3 convolutions (unpadded convolutions), each followed by a rectified linear unit (ReLU) and a 2x2 max pooling operation with stride 2 for downsampling. At each downsampling step we double the number of feature channels. Every step in the expansive path consists of an upsampling of the feature map followed by a 2x2 convolution (“up-convolution”) that halves the number of feature channels, a concatenation with the correspondingly cropped feature map from the contracting path, and two 3x3 convolutions, each followed by a ReLU. The cropping is necessary due to the loss of border pixels in every convolution. At the final layer a 1x1 convolution is used to map

each 64-component feature vector to the desired number of classes. In total the network has 23 convolutional layers.

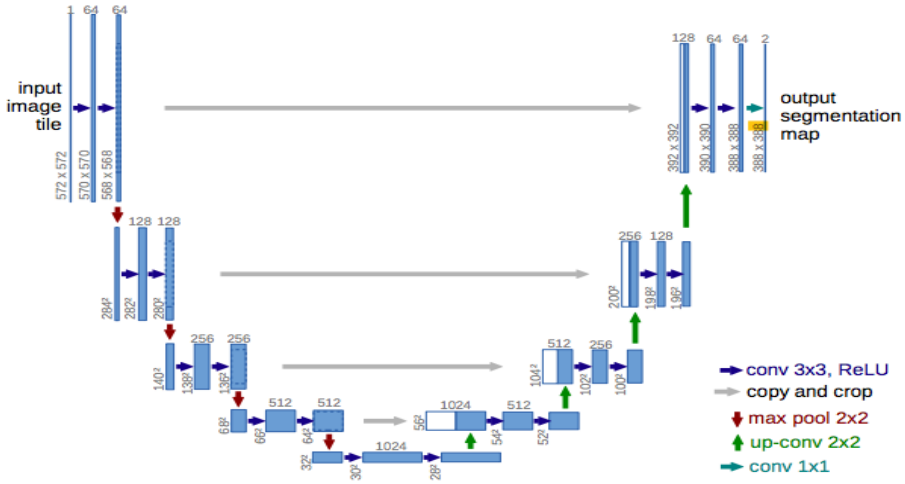


Fig: 2.5- Architecture of UNet

3.4 PROPOSED SYSTEM IMPLEMENTATION:

There are three main stages: pre-processing, classification via CNN and post-processing.

A. Pre-Processing:

MRI images are altered by the bias field distortion. This makes the intensity of the same tissues to vary across the image. To correct it, we applied the N4ITK method. However, this is not enough to ensure that the intensity distribution of a tissue type is in a similar intensity scale across different subjects for the same MRI sequence, which is an explicit or implicit assumption in most segmentation methods. In fact, it can vary even if the image of the same patient is acquired in the same scanner in different time points, or in the presence of pathology. So, to make the contrast and intensity ranges more similar across patients and acquisitions, we apply the intensity normalization method proposed by each sequence. In this intensity normalization method, a set of intensity landmarks are learned for each

sequence from the training set. And are chosen for each MRI sequence as described in, represents the intensity at the percentile. After training, the intensity normalization is accomplished by linearly transforming the original intensities between two landmarks into the corresponding learned landmarks. In this way, the histogram of each sequence is more similar across subjects. After normalizing the MRI images, we compute the mean intensity value and standard deviation across all training patches extracted for each sequence. Then, we normalize the patches on each sequence to have zero mean and unit variance1.

B. Convolutional Neural Network CNN:

CNN was used to achieve some breakthrough results and win well-known contests. The application of convolutional layers consists in convolving a signal or an image with kernels to obtain feature maps. So, a unit in a feature map is connected to the previous layer through the weights of the kernels. The weights of the kernels are adapted during the training phase by back propagation, in order to enhance certain characteristics of the input. Since the kernels are shared among all units of the same feature maps, convolutional layers have fewer weights to train than dense FC layers, making CNN easier to train and less prone to overfitting. Moreover, since the same kernel is convolved over the entire image, the same feature is detected independently of the location—translation invariance. By using kernels, information of the neighborhood is taken into account, which is an useful source of context information. Usually, a non-linear activation function is applied on the output of each neural unit. If we stack several convolutional layers, the extracted features become more abstract with the increasing depth. The first layers enhance features such as edges, which are aggregated in the following layers as motifs, parts, or objects. The following concepts are important in the context of CNN:

1) Initialization: It is important to achieve convergence. We use the Xavier initialization. With this, the activations and the gradients are maintained in controlled levels, otherwise back-propagated gradients could vanish or explode.

2) Activation Function: It is responsible for non-linearly transforming the data. Rectifier linear units (ReLU), defined as were found to achieve better results than the more classical sigmoid, or hyperbolic tangent functions, and speed up training. However, imposing a constant 0 can impair the gradient flowing and consequent adjustment of the weights. We cope with these limitations using a variant called leaky rectifier linear unit (ReLU) that introduces a small slope on the negative part of the function.

3) Pooling: It combines spatially nearby features in the feature maps. This combination of possibly redundant features makes the representation more compact and invariant to small image changes, such as insignificant details; it also decreases the computational load of the next stages. To join features it is more common to use max-pooling.

4) Regularization: It is used to reduce overfitting. We use Dropout, in the FC layers. In each training step, it removes nodes from the network with probability .In this way; it forces all nodes of the FClayers to learn better representations of the data, preventing nodes from co-adapting to each other. At test time, all nodes are used. Dropout can be seen as an ensemble of different networks and a form of bagging, since each network is trained with a portion of the training data.

5) Data Augmentation: It can be used to increase the size of training sets and reduce over fitting. Since the class of the patch is obtained by the central voxel, we restricted the data augmentation to obtaining operations. Some authors also consider image translations, but for segmentation this could result in attributing a wrong class to the patch. So, we increased our data set during training by generating new patches through the rotation of the

original patch. In our proposal, we used angles multiple of 90, although another alternative will be evaluated.

6) Loss Function: It is the function to be minimized during training. We used the Categorical Cross-entropy, where represents the probabilistic predictions (after the softmax) and is the target. In the next subsections, we discuss the architecture and training of our CNN.

7) Architecture: We aim at a reliable segmentation method; however, brain tumors present large variability in intra-tumoral structures, which makes the segmentation a challenging problem. To reduce such complexity, we designed a CNN and tuned the intensity normalization transformation for each tumor grade—LGG and HGG. The proposed architectures are presented in Tables I and II2. The architecture used for HGG is deeper than the one for LGG, because going deeper did not improve results in the latter. To go deeper, one must include more layers with weights, which may increase overfitting, given the smaller training set of LGG. This is supported by the need of setting Dropout with in LGG, while it is in HGG, since the database used for evaluation contained more HGG then LGG cases. Additionally, the appearance and patterns are different in HGG and LGG. Since we are doing segmentation, we need a precise sense of location. Pooling can be positive to achieve invariance and to eliminate irrelevant details; however, it can also have a negative effect by eliminating important details. We apply overlapping pooling with 3*3 receptive fields and 2*2 stride to keep more information of location. In the convolutional layers the feature maps are padded before convolution, so that the resulting feature maps could maintain the same dimensions. In the case of HGG there are 2,118,213 weights to train, while in LGG it lowers to 1, 933, 701 weights because it has two less convolutional layers. All sequences were used as input. ReLU is the activation function in all layers with weights, with the exception of the last that uses softmax. Drop out was used only in the FClayers.

8) Training: To train the CNN the loss function must be minimized, but it is highly non-linear. We use Stochastic Gradient Descent as an optimization algorithm, which takes steps proportionally to the negative of the gradient in the direction of local minima. Nevertheless, in regions of low curvature it can be slow. So, we also use Nesterov's Accelerated Momentum to accelerate the algorithm in those regions. The momentum is kept constant, while the learning rate was linearly decreased, after each epoch. We consider an epoch as a complete pass over all the training samples.

C. Post-processing:

Some small clusters may be erroneously classified as tumor. To deal with that, we impose volumetric constrains by removing clusters in the segmentation obtained by the CNN that are smaller than a predefined threshold.

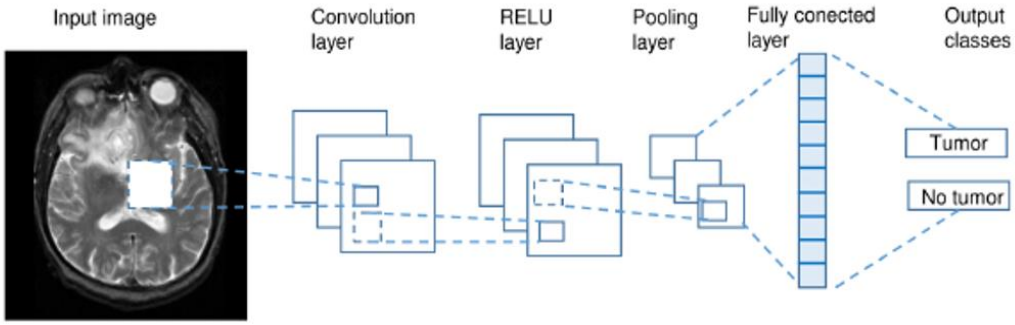


Fig: 2.6- Flow of the project

3.5 ARCHITECTURE OF PROPOSED SYSTEM:

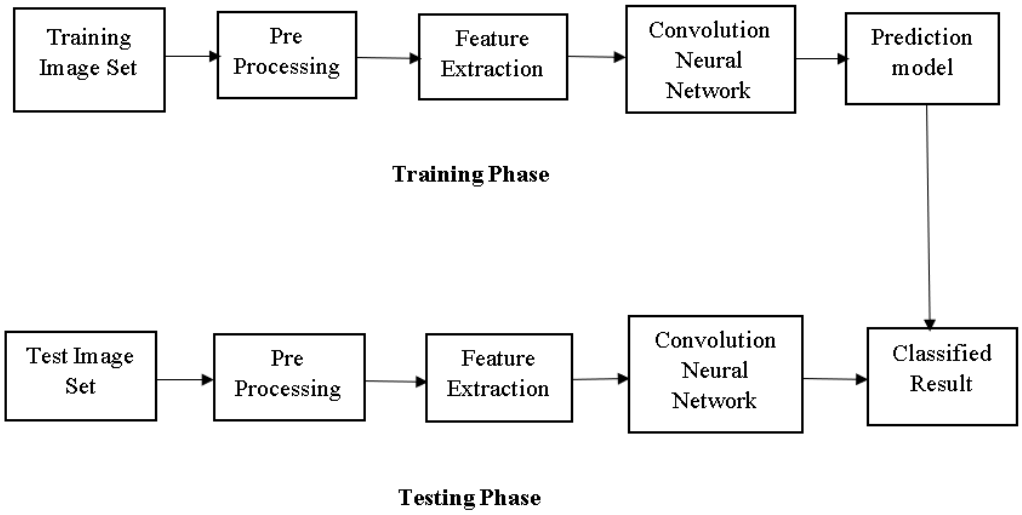


Fig: 2.7- Architecture of project

3.6 HARDWARE AND SOFTWARE REQUIREMENTS:

3.6.1 HARDWARE REQUIREMENTS:

- Processor : i3 & any latest versions.
- Hard Disk : 500 GB.
- Monitor : 15'' LED.
- Input Devices : Keyboard, Mouse.
- RAM : 4 GB.

3.6.2 SOFTWARE REQUIREMENTS:

- Programming Language / Platform : Python
- IDE : pycharm
- Web Framework : Django
- UI Technologies : html, css, js
- Database : sqlite

CHAPTER-4

DESIGNING

4.1 UML DIAGRAMS:

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.. The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

Do we really need UML?

Complex applications need collaboration and planning from multiple teams and hence require a clear and concise way to communicate amongst them.

Businessmen do not understand code. So UML becomes essential to communicate with non-programmer’s essential requirements, functionalities and processes of the system. A lot of time is saved down the line when teams are able to visualize processes, user interactions and static structure of the system. UML is linked with object oriented design and analysis. UML makes the use of elements and forms associations between them to form diagrams. Diagrams in UML can be broadly classified as:

The Primary goals in the design of the UML are as follows:

Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models. Provide extendibility and specialization mechanisms to extend the core concepts. Be independent of particular programming languages and development process. Provide a formal basis for understanding the modeling language. Encourage the growth of OO tools market. Support higher level development concepts such as collaborations, frameworks, patterns and components. Integrate best practices.

Types of UML Diagrams:

1. Structural Diagrams:

Capture static aspects or structure of a system. Structural Diagrams include: Component Diagrams, Object Diagrams, Class Diagrams and Deployment Diagrams.

2. Behavior Diagrams:

Capture dynamic aspects or behavior of the system. Behavior diagrams include: Use Case Diagrams, State Diagrams, Activity Diagrams and Interaction Diagrams.

The image below shows the hierarchy of diagrams according to UML

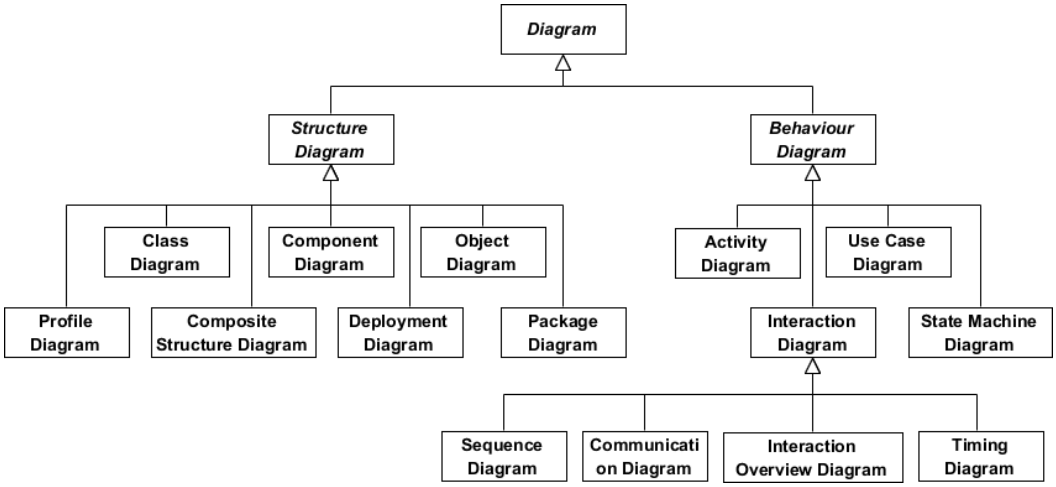


Fig: 2.8- Types of UML Diagram

4.2 USECASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

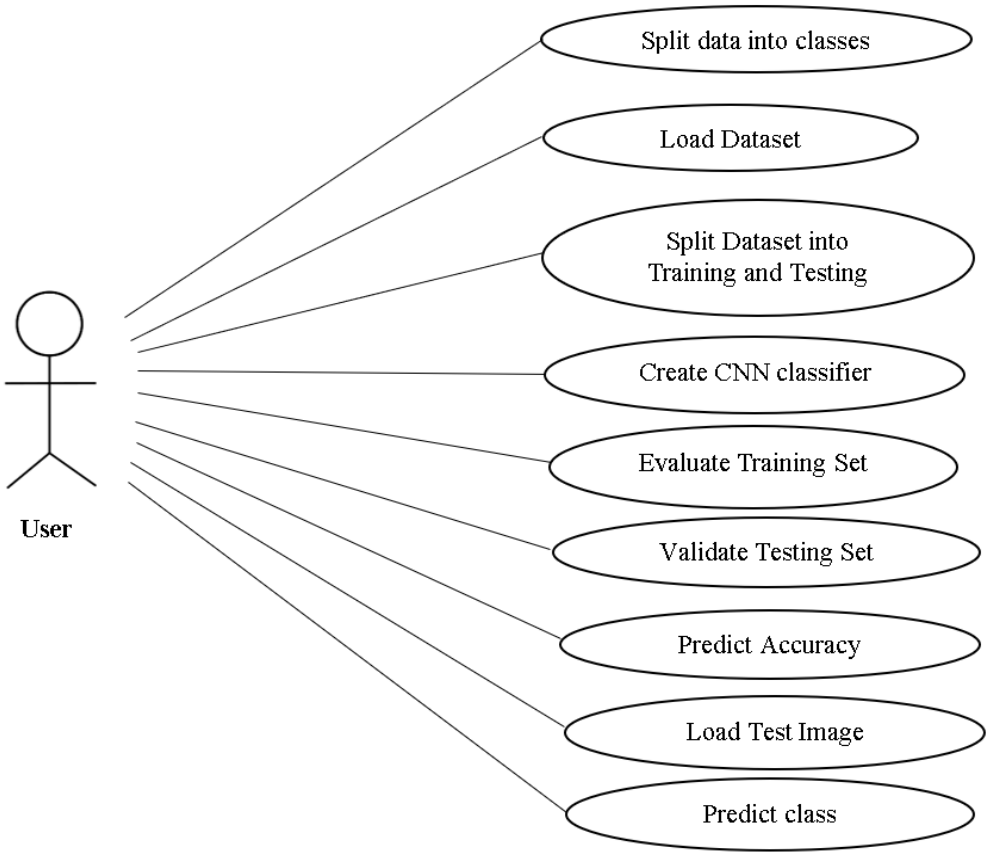


Fig: 2.9- Usecase Diagram

4.3 SEQUENCE DIAGRAM:

Sequence Diagrams Represent the objects participating the interaction horizontally and time vertically. A Use Case is a kind of behavioral classifier that represents a declaration of an offered behavior. Each use case specifies some behavior, possibly including variants that the subject can perform in collaboration with one or more actors. Use cases define the offered behavior of the subject without reference to its internal structure. These behaviors, involving interactions between the actor and the subject, may result in changes to the state of the subject and communications with its environment.

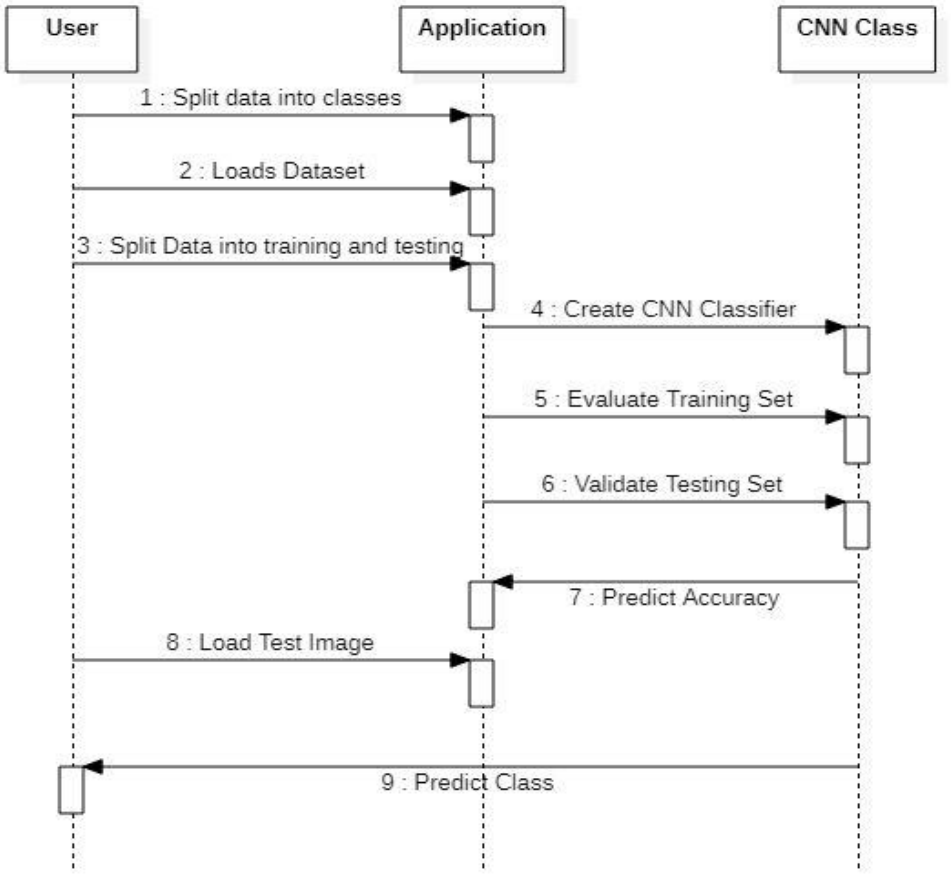


Fig: 3.0- Sequence Diagram

4.4 ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of Workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

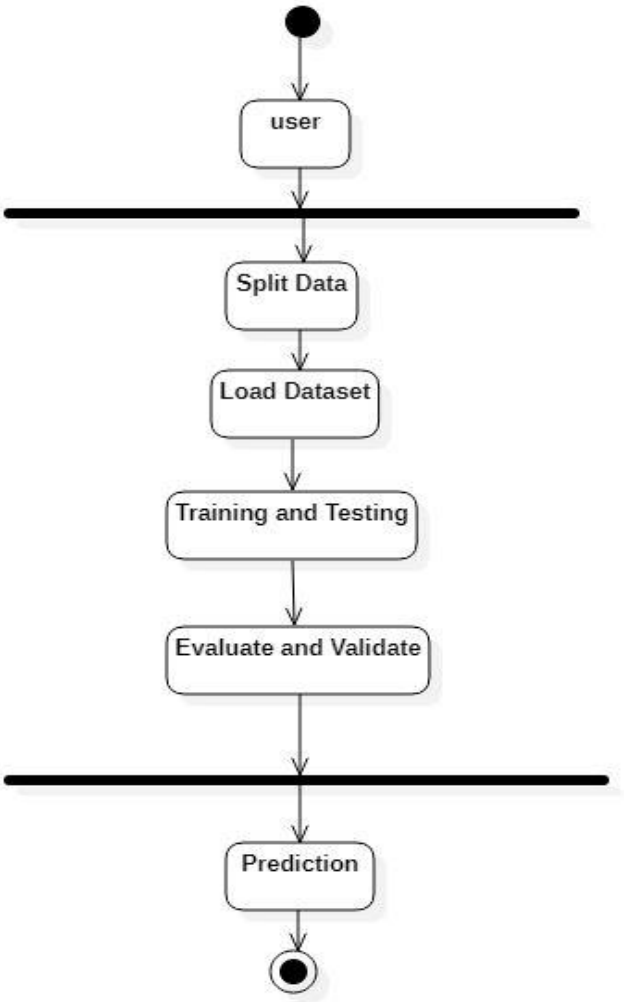


Fig: 3.1- Activity Diagram

4.5 CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

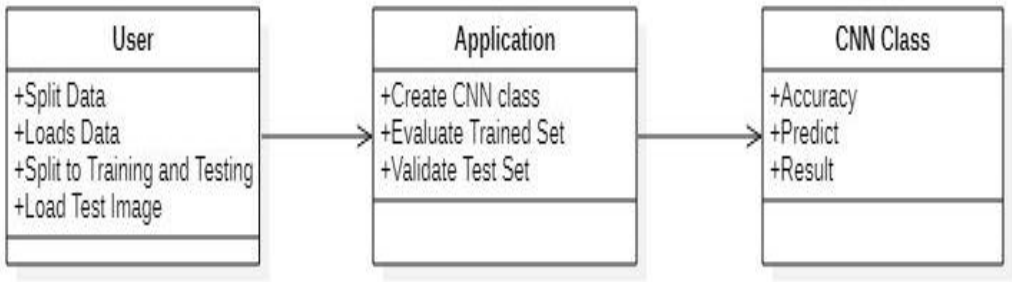


Fig: 3.2- Class Diagram

CHAPTER-5

RESULT AND DISCUSSION

5.1 PERFORMANCE ANALYSIS:

```
In [4]: plt.figure(figsize=(12, 5))
        for i, idx in enumerate(np.random.randint(images.shape[0], size=18), start=1):
            plt.subplot(3, 6, i)
            plt.imshow( np.squeeze(images[idx],axis=-1), cmap='gray')
            plt.imshow( np.squeeze(np.ones_like(masks[idx])-masks[idx],axis=-1), alpha=0.5, cmap='Set1')
            plt.title(integer_to_class[str(labels[idx])])
            plt.axis('off')
```

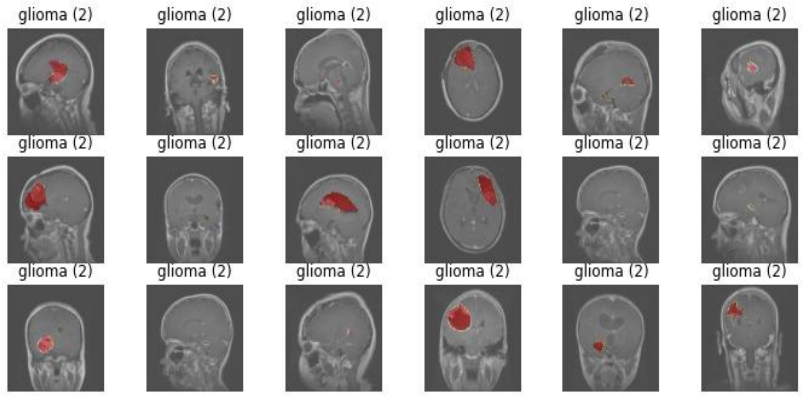


Fig: 3.3- Training the MRI images

Train Image and its mask which is to be predicted

```
In [5]: plt.figure(figsize=(12, 5))
        i=1
        for idx in np.random.randint( images.shape[0], size=9):
          plt.subplot(3,6,i);i+=1
          plt.imshow( np.squeeze(images[idx],axis=-1))
          plt.title("Train Image")
          plt.axis('off')
          plt.subplot(3,6,i);i+=1
          plt.imshow( np.squeeze(masks[idx],axis=-1))
          plt.title("Train Mask")
          plt.axis('off')
```

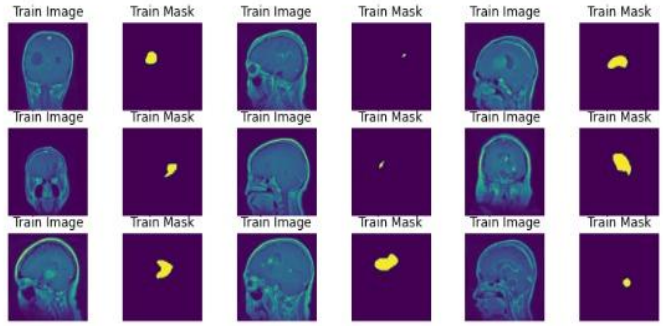


Fig: 3.4- Masking images

The actual purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner.

Dice Loss:

Dice loss function is used for image segmentation tasks is based on the Dice coefficient, which is essentially a measure of overlap between two samples. This measure ranges from 0 to 1 where a Dice coefficient of 1 denotes perfect and complete overlap. The Dice coefficient was originally developed for binary data, and can be calculated as:

$$Dice = \frac{2\,|A \cap B|}{|A| + |B|} \qquad DSC = \frac{2TP}{2TP + FP + FN}.$$

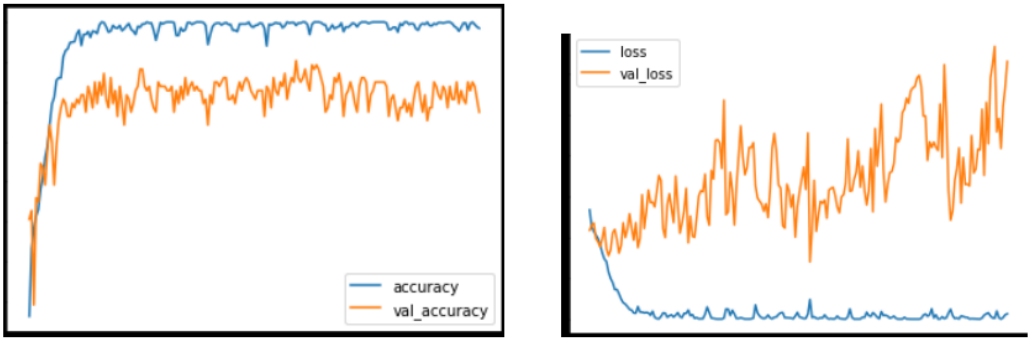


Fig: 3.5- Comparing Accuracy and Loss of Model

5.3 OUTPUT SCREEN SHORTS:

- Firstly, we have to paste the URL: 127.0.0.1:8000, in the google chrome, then we'll get the above view.

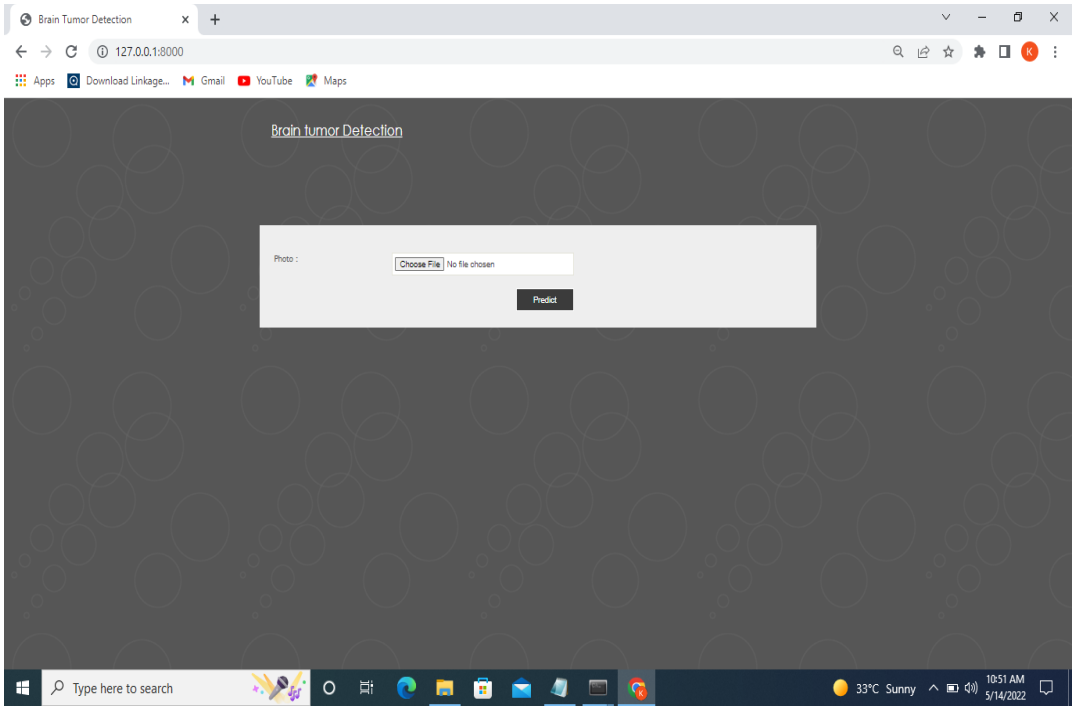


Fig: 3.6- Image of the Web page

- By clicking the Choose File button it will open the MRI images folder, then we can choose any of the MRI image, and then we have to click open button. Then above two figures shows how we select the wanted MRI image. And it also shows which file we have selected with its file name. Then if we press predict button it shows the final output image.

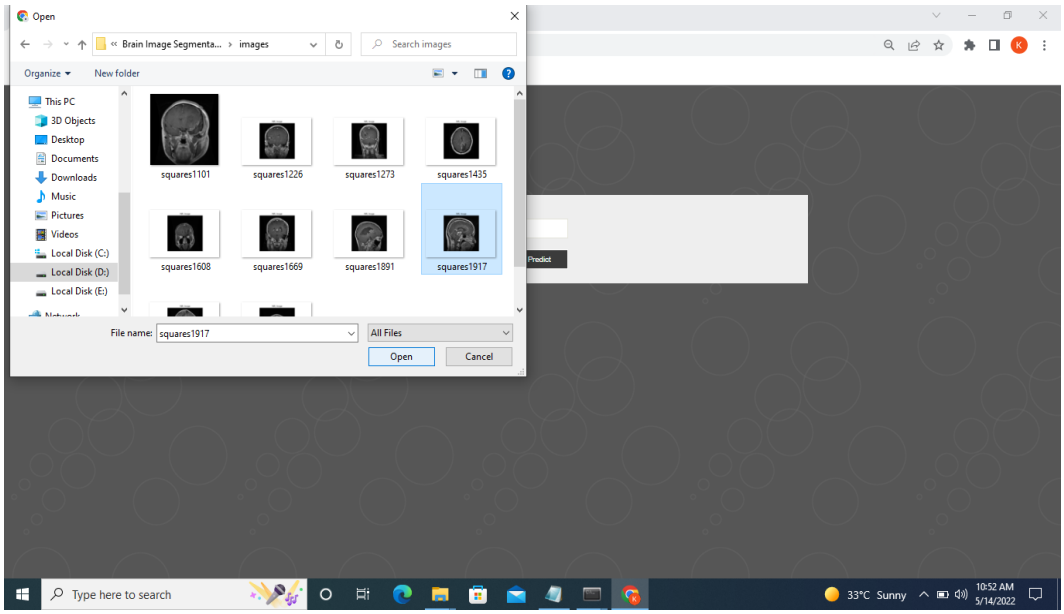


Fig: 3.7- Image showing how to choose MRI Image

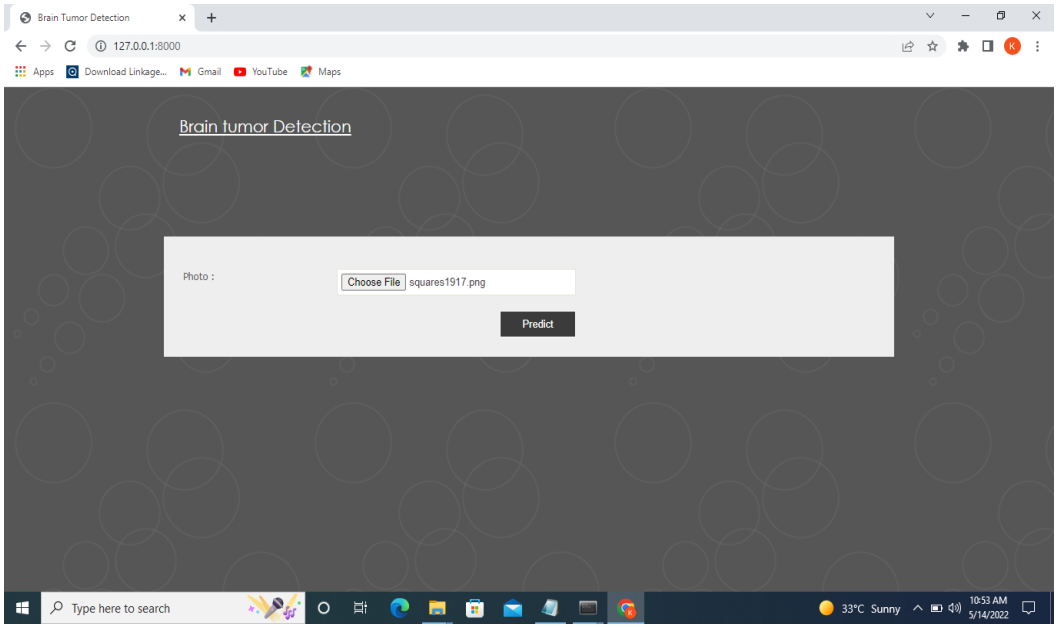


Fig: 3.8- Image showing the file name

- After pressing the predict button then we'll get the predicted mask output with its tumor location along with its dimensions.

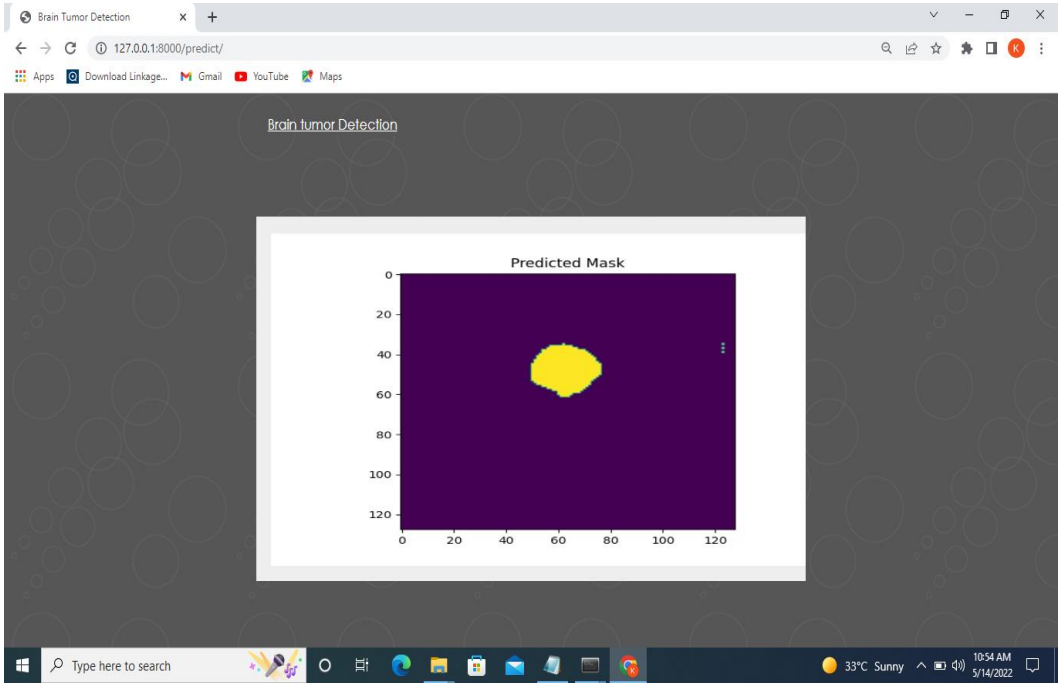


Fig: 3.9- Result of Predicted mask image

CHAPTER-6

SUMMARY AND CONCLUSION

5.1 SUMMARY:

Among brain tumors, gliomas are the most common and aggressive, leading to a very short life expectancy in their highest grade. Thus, treatment planning is a key stage to improve the quality of life of oncological patients. Magnetic resonance imaging (MRI) is a widely used imaging technique to assess these tumors, but the large amount of data produced by MRI prevents manual segmentation in a reasonable time, limiting the use of precise quantitative measurements in the clinical practice. So, automatic and reliable segmentation methods are required; however, the large spatial and structural variability among brain tumors make automatic segmentation a challenging problem. In this paper, we propose an automatic segmentation method based on Convolutional Neural Networks (CNN), exploring small 3*3 kernels. We also investigated the use of intensity normalization as a pre-processing step, which though not common in CNN-based segmentation methods, proved together with data augmentation to be very effective for brain tumor segmentation in MRI images.

5.2 CONCLUSION:

For the segmentation of brain tumors in MRI images, we offer a new CNN-based technique. Pre-processing, CNN classification, and post-processing are the three key processes. We compute the mean intensity value and standard deviation across all training patches retrieved for each sequence after normalizing the MRI images. The patches on each sequence are then normalized to have a zero mean and unit variance. The CNN design was used to segment the brain tumor. Because it aids in proper segmentation, the algorithm connects both local and global features. Using max pooling, max out, and drop out to supplement the learning process, the training and testing speed is boosted. The fully linked layer's speed is improved by lowering the features. Reduced parameter values also result in overfitting. The result shows that the implemented method helps in detection of enhancing tumor as well as specifying tumor to the actual tumor region only.

CHAPTER-7

REFERENCES

[1] Disha Sharma, Gagandeep Jindal, “Identifying Brain Cancer Using Image Processing Techniques”, International Conference on Computational Techniques and Artificial Intelligence, pp. 116-120, 2011.

[2] Chaudhary and S. S. Singh, "Brain Cancer Detection on CT Images by Using Image Processing," 2012 International Conference on Computing Sciences, Phagwara, 2012, pp. 142-146.

[3] Amutha and R. S. D. Wahidabanu, "Brain tumor detection and diagnosis in CT scan images," 2013 International Conference on Communication and Signal Processing, Melmaruvathur, 2013, pp. 1108-1112.

[4] Prof. Anuradha S. Deshpande, Dhanesh D. Lokhande, “Brain Cancer Detection with fusion of CT and MRI images using image processing”, International Journal of advanced research in computer engineering and technology (IRARCET), Vol.4 Issue 3, March 2015.

[5] Sijbers J et al (1998), “Estimation of the noise in magnitude MR images”, Magn Reson Imaging 16(1):87- 90.

[6] MathWorks(https://in.mathworks.com/help/wavelet/ug/lifting-method-forconstructingwavelets.html).